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PERSPECTIVE

# Artificial intelligence, medications, pharmacogenomics, and ethics

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**ABSTRACT**
Artificial Intelligence (AI) and Machine Learning (ML) are revolutionizing various scientific and clinical disciplines including pharmacogenomics (PGx) by enabling the analysis of complex datasets and the development of predictive models. The integration of AI and ML with PGx has the potential to provide more precise, data-driven insights into new drug targets, drug efficacy, drug selection, and risk of adverse events. While significant effort to develop and validate these tools remain, ongoing advancements in AI technologies, coupled with improvements in data quality and depth is anticipated to drive the transition of these tools into clinical practice and delivery of individualized treatments and improved patient outcomes. The successful development and integration of AI-assisted PGx tools will require careful consideration of ethical, legal, and social issues (ELSI) in research and clinical practice. This paper explores the intersection of PGx with AI, highlighting current research and potential clinical applications, and ELSI including privacy, oversight, patient and provider knowledge and acceptance, and the impact on patient-provider relationship and new roles.

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## 1. Introduction

Pharmacogenomics (PGx) is an integral part of personalized medicine, aiming to optimize drug efficacy and minimize adverse drug reactions (ADRs) by incorporating genetic factors into treatment decisions. Several single genes have been found to cause low or non-efficacy or ADRs and PGx testing is available. Over the years, various traditional research approaches including biochemical assays, familial studies, animal models and human studies have led to the identification of several causative pharmacogenes and their variations. With the advent of a wide range of molecular tools in the 1990s through today, the rate of discovery accelerated the pace of PGx research and led to development of new clinical tests.

However, while poor efficacy or ADRs have been associated with variants in single genes, suggesting a Mendelian inheritance with high penetrance, it is clear that drug response is a complex phenotype with many factors impacting efficacy or risk of ADRs. With various -omic technologies, more comprehensive analysis can be performed of DNA variants in combination with functional assays. Assessing a single gene at a time was replaced by broad-scale analysis of multiple genes or the entire genome. The addition of non-genomic datasets adds yet another layer of complexity, critical to capturing unique patient variables, but poses substantial analytical challenges.

Artificial intelligence (AI) is a broad field that has enabled the analysis of substantially large datasets to identify patterns associated with particular phenotypes. Dating back to the 1940s [1], it has become increasingly useful in data sciences with large genomic and clinical datasets. In particular, such analytical tasks include reasoning, problem-solving, understanding natural language, perception, and decision-making. Machine learning (ML) is a subset of AI that focuses on developing algorithms that enable predictions based on data. There are supervised and unsupervised methods in ML (outputs missing in unsupervised methods). These algorithms improve automatically through experience. DL is a subset of ML that uses neural networks with many layers (deep neural networks) to model complex patterns in large datasets. A subset of ML, deep learning (DL) models, especially those involving convolutional neural networks (CNNs) and recurrent neural networks (RNNs), have shown great promise in tasks that involve high-dimensional data.

Though perhaps not as advanced as other fields such as radiology [2–5], AI and ML techniques have been increasingly applied in PGx research to analyze genomic data, predict drug responses, and identify potential genetic markers for drug metabolism and efficacy [6–9]. With the expanding scope of data generated from new genomic technologies, other clinical biomarkers, and patient factors such as polypharmacy and co-morbidities (data that can be abstracted from the electronic medical records (EMRs), AI and ML can facilitate the integration and analysis of such large and diverse datasets from different sources [10]. This holistic approach enables a much more comprehensive understanding of both the genetic and environmental (and their interaction) causes of drug response and efficacy.

The intersection of PGx and AI represents a transformative approach to precision medicine to tailor treatments and improve patient outcomes [6,11–16]. While these anticipated advancements can improve therapeutic decision-making and outcomes, the development and use of these tools has raised ethical, legal, and social issues (ELSI) of concern in healthcare.

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**Article highlights**

- This article provides a brief overview of current developments in AI as it relates to development and use of medications, particularly PGx data, as well as key ELSI.
- AI-PGx raises several ELSI including patient privacy, patient and provider knowledge and acceptance, and the impact on patient-provider relationship and new roles.
- As the scope and depth of clinical applications utilizing AI continue to grow, it is important to consider and address ELSI that may impede the development and clinical utilization of these tools.
- Appropriate utilization of AI-PGx tools require that both providers and patients understand how they will be integrated into clinical care and what protections are in place.

This perspective aims to highlight key uses of AI in PGx research and future clinical applications.

## 2. Use of AI/ML in medications & PGx

While the use of AI/ML is still relatively early in the field of PGx, there are many areas related to drug development, safety and efficacy that are developing and evaluating AI-based tools. This section includes some brief descriptions of some of these developments and applications. There are several review articles written AI applications in medicine that may provide a more in-depth discussion of the challenges related to the methods [17–19].

### *2.1. Predicting drug response & optimizing drug selection*

Predictive models using ML algorithms, such as random forests, support vector machines, and neural networks, have shown promise in predicting individual drug responses based on genetic profiles enabling recommendations for safer treatment alternatives or dosage adjustments [20–23]. For example, ML models have been developed to predict warfarin dosage requirements based on genetic variants and other clinical variables [24–29]. The ability to distinguish between responders and non-responders has benefitted from larger datasets and ML/DL tools. For instance, DL algorithms have been employed to predict drug-gene interactions by integrating multi-omics data in cancer [30]. ML approaches have been evaluated to predict responses to treatments for mental health disorders based on clinical data such as baseline severity and PGx variants [10,31], prior treatment response [32,33], and PGx data in combination with demographic data [34], or metabolomic and variant data [35]. However, some work has indicated that algorithms utilizing variant databases and rules of pathogenicity may have sub-par performance for PGx variants [36]. With respect to optimizing drug selection, several AI tools are under development to optimize drug selection and response in oncology, for example [37–40].

In addition, AI can play a role in post-market surveillance through analysis of data from various sources, including social media, patient forums, and adverse event reporting systems, to detect signals of previously unrecognized adverse effects. This post-market surveillance can assist in the early identification of drug safety issues that might not have been evident during clinical trials or through traditional reporting systems. AI can harness real-world data from patient registries, EMRs, and other sources to provide insights into the long-term safety and efficacy of drugs.

### *2.2. Drug target discovery & design*

AI algorithms can help identify candidate drug compounds for certain diseases through analysis of multiple chemical and biological datasets [41] and potentially increase the pace of discovery of new therapeutic targets [42–46]. AI algorithms have already identified novel genetic markers linked to response to cancer therapies, aiding in the development of targeted treatments [47–49]. Particularly for less well-understood conditions with limited therapeutic options, the identification of new targets has been enabled by ML and DL algorithms, particularly based on transcriptomic datasets. For example, Zeng et al. identified four potential targets for lupus nephritis based on differentially expressed gene sets [50]. Similarly, other studies have used RNA analysis to identify targets for acute type A aortic dissection [51] and major depression disorder [52]. These data can be used in *in vitro* experimental models to assess potential effects of protein inhibition or other mechanisms. Furthermore, AI tools are being developed for molecule generation based on chemical language models [53,54].

With the increasing number of targeted therapies available, ascertainment of a patient's genetic make-up is necessary. AI predictive models can be used to determine mutation status based on several clinical factors including imaging data [55,56].

### *2.3. Prediction of drug-drug interactions*

Drug-drug interactions (DDI) contribute to many adverse responses and accounting for a number of hospital admissions [57]. AI systems can be used to predict and continuously monitor a patient's medication regimen and flag potential drug-drug interactions [58–61], extracting relevant information from clinical notes and published literature to provide real-time alerts to healthcare providers. For example, an artificial neural network has been developed and evaluated to predict vancomycin-induced nephrotoxicity, which can be caused by concomitant use of piperacillin – tazobactam and vasopressor drugs [20]. Another application utilizes a DL model to predict DDI in patients with multiple sclerosis [62].

In addition, diet, smoking and other factors may influence the metabolism of medications and result in adverse reactions. New ML predictive models are being developed and evaluated to explore these interactions, potentially providing more tailored recommendations for patients to optimize drug safety and efficacy [63].

### *2.4. Clinical applications & clinical decision support (CDS)*

The numerous factors that impact ADR risk and therapeutic outcome pose challenges for treatment decision-making at the point of care in various settings. The benefits of AI-driven CDS tools span the clinical spectrum from diagnosis to treatment to predicting recurrence and minimize provider variation, reducing errors, and gathering pertinent clinical data from the medical record [14,64]. For example, these tools can alert providers to potential safety concerns such as allergies, contraindications, and optimal dosing, enhancing the safety of drug prescribing and administration. AI has been used for some time for clinical applications for screening [65], diagnosis and predicting outcomes [66,67]; and genomic data is increasingly included in these tools [68]. AI tools have been under development to optimize treatment decisions through various CDS tools in a number of specialties [69]. AI tools based on non-genomic datasets have been developed to optimize dosing and drug selection [70–73]. For example, AI tools have been developed to help improve diagnosis and inform treatment decisions for acute ischemic stroke [74] and corneal diseases [75]. AI-based tools for drug prediction and sensitivity based on gene expression data and disease-specific signaling networks could inform cancer patients' optimal treatment regimen [76]. Although not yet including PGx variants, intelligent prescription systems (IPS) can assess several patient variables to generate optimal prescription recommendations, greatly assisting prescribers [77].

AI-driven CDS can also integrate PGx data into clinical workflows, providing clinicians with actionable recommendations about medication selection and dosing tailored to a patient's genetic profile, lifestyle data, and other health metrics [78]. Disparities between and within clinical practices can be reduced or eliminated through the use of AI-driven CDS tools with respect to diagnosis and clinical decisions.

Related to genetic testing and the challenges of variant interpretation, a number of AI-driven applications have been developed to extract data from multiple databases for disease-based [79–81] and PGx variant interpretation [82–84]. Molecular testing or pathology laboratories' utilization of AI-assisted results interpretation of PGx [84–86] and recommendations based on CPIC or other guidelines have been shown to be effective [13].

In addition to provider tools, AI-supported tools are being developed for patients. In particular, chatbots have been developed to foster patient education of complex health issues [87], a challenge that is particularly relevant in genetics. For PGx, an AI assistant was developed using OpenAI's GPT-4 and found to be responsive to patients' questions [13]. A voice-activated AI tool was reported to be successful in significantly optimizing insulin dosing for Type II diabetic patients compared to traditional dosing approaches [88]. AI tools have also been developed for patient support for making health behavior changes, stress, weight loss, and disease management [89–92].

Another type of patient-supporting AI-driven application is related to monitoring medication behavior. Poor adherence is a major challenge for many chronic medications, leading to lower than desired benefits and poorer health outcomes. ML has been applied to several tools to assess and/or promote medication adherence [93,94]. In particular, applications using wearable sensors that utilize ML have been developed and evaluated [95–97]. These tools can provide real-time data to healthcare providers, enabling timely interventions and improving overall drug safety.

## 3. Ethical, legal, and societal implications for AI-PGx research and clinical applications

Despite the potential benefits of AI in drug development, improving the safe use of drugs, and optimizing response, several challenges may impede the widespread and ethical adoption of AI in PGx. A recent review reported that while there is general support for AI in medicine, several factors may reduce utilization, though responses are unique to specific clinical scenarios [98]. According to a 2023 Pew Research Center survey, 60% of Americans would be uncomfortable if their health provider relied on AI for their healthcare decisions [99]. The majority of respondents (71%) wouldn't use a chatbot for mental health support [99]. However, these attitudes are not consistent between populations. For example, one survey study reported that African-American respondents were less likely to select AI but Native Americans were more likely to select it diverse patients [100].

The development of AI-based CDS tools can help rapidly define the best course of treatment based on the PGx test results and other patient variables. While no studies have specifically explored attitudes toward AI for PGx testing or therapeutic decision-making, studies have examined attitudes toward AI tools for prescribing. For example, Rawson et al [101]. reported patients were less likely to use a microneedle biosensor and automated antibiotic dosing system versus a traditional blood draw. Specifically, while they supported use of the microneedle technology, they preferred for a health provider to decide the appropriate antibiotic dosing. For PGx in particular, therapeutic recommendations based on the patient's genotype, medication history, co-morbidities, polypharmacy, and patient preferences may conflict with AI-generated recommendations.

With the rapid development and use of ML/AI tools for health and non-health applications, several groups have begun to develop ethical practice guidelines and frameworks for AI [102–107]. An overarching framework can help identify and guide responsible implementation of ML/AI tools in clinical practice, although the nuances of different clinical AI tools including PGx may require different approaches. For example, Char (2020) proposed a 4-staged framework of ethical issues related to ML tools in healthcare: 1) development and implementation; 2) regulation and oversight; 3) value-based issues related to tool characteristics or decisions; and 4) ethical issues [108]. In the following section, I briefly discuss several issues in these areas and highlight issues relevant to AI-PGx applications. Many of these issues overlap between research and anticipated clinical applications and are relevant to the broader genetics/genomics field.

### 3.1. *Privacy and genetic data*

One major issue regarding development and training of AI tools is the privacy of patient health data [109–112]. Ensuring the confidentiality and secure handling of genetic information is paramount. Genetic data is inherently sensitive, revealing not only information about an individual but also insights into familial and ancestral connections, making true anonymization challenging. Privacy of genetic data has been a long-standing concern for both research participants and patients, leading to development of special federal protections in 2008 [113]. As AI tools require vast datasets to train accurate models, genetic data often must be shared across institutions, increasing the risk of data breaches and unauthorized access. Furthermore, AI's pattern-recognition capabilities can amplify the risk of re-identifying individuals from de-identified genetic data by cross-referencing with other datasets, such as public genealogy databases. Therefore, if patient genetic data are to be used for the development of AI tools, maximizing patient privacy is critical and the need for consent should be considered. An institution's governance policies regarding sharing of patient data for development of AI tools or other clinical applications should be transparent and readily accessible to patients. Several encryption tools are available to increase privacy of patient data [111,114,115]. Federated learning, which involves training models across multiple decentralized datasets while preserving data privacy, may help mitigate biased tools for PGx. This approach can overcome data-sharing limitations, protect patient privacy, and enhance the robustness of AI models by leveraging diverse datasets [116,117]. The sharing of genetic testing results may increase patient concerns, though many participants of large genetic and genomic biobanks are willing to share their de-identified or anonymous data [118]. Without strict data governance protocols, there is a risk that genetic information may be misused, leading to genetic discrimination in areas like insurance or employment, despite legal protections like the Genetic Information Nondiscrimination Act (GINA).

### *3.2. Patient confidence/trust in AI-PGx tools*

AI applications are likely to be novel to both patients and providers alike, though many people are exposed to these tools through their use by retailers, news media, and other common public entities. In addition to concerns regarding privacy and confidentiality, many factors may impact patient acceptability and trust of AI tools, including the specific application and/or type of health question/issue. Some studies have reported patient and provider optimism in the use of AI tools and appreciation of its benefits in the clinical setting, such as increased accuracy and rapidity of interpretation of images or test results [119]. For example, for a screening intervention for diabetic retinopathy, patients were highly satisfied with the AI tool compared to the manual screening tool and most actually preferred the AI tool [120]. But other studies have reported that even if an AI tool is shown to outperform a human-based diagnosis, patients may still prefer the human-based diagnosis [121]. With respect to information accuracy, some studies have reported high accuracy of AI-generated responses to common patient questions [122–125], while other reports have raised concerns about response accuracy for medically-related questions [126]. If the information is incomplete or inaccurate, patient harms may occur. Trust in AI tools may be influenced by the recommendation of healthcare providers. When providers demonstrate confidence in AI and integrate it effectively into their practice, patients are more likely to accept and trust these tools. In addition, patient comfort and accessibility may impact utilization of AI resources within the clinic and outside.

Furthermore, patient understanding about how information from AI tools will be used in healthcare decisions will be critical. For PGx-guided treatments, reduced confidence in test results or treatment recommendations could adversely impact adherence and result in less-than-optimal health outcomes. If AI tools are believed to supplement, rather than replace, a health care provider's decision-making, it may be accepted more. With respect to the interpretation of PGx test results and applying them to therapeutic decisions, some patient factors should be incorporated into an AI-driven model and thus, may require adjustment of the AI-generated recommendation to integrate patient variables.

During this early phase of development and clinical implementation, patient education and communication will be essential to the appropriate utilization of AI tools. Clear explanations from healthcare providers about how AI tools work, the data it uses, and its limitations can enhance acceptability. Explainable AI (XAI) models that provide interpretable outputs can help bridge the gap between complex algorithms and patient understanding. More transparency about the accuracy and limitations of these tools should be readily available to patients to inform appropriate use and promote understanding [127]. Consent, such as an opt-in approach, may be warranted to respect patient preferences and comfort. However, truly informed consent will require understanding of the performance characteristics and dataset used to train the tool. With increased awareness and integration of AI tools in medicine, consent may no longer be required or the approach can shift to an opt-out model.

### *3.3. Provider willingness/acceptance/readiness*

As with patients, providers are likely to display differences in uptake and acceptance of AI tools in practice. Provider behaviors can be impacted by several factors [128–131], including medical liability [132,133], how AI tools are integrated into the clinical workflow [134], uncertain or unclear evidence basis [135,136], and clinical decision-making control [135]. With respect to liability, providers are ultimately responsible for patient outcomes; relying on AI recommendations introduces questions of accountability if an AI-driven decision leads to harm. Thus, some providers may avoid AI to reduce liability risks, preferring to rely on their own expertise.

Another issue is related to the clinical workflow. Integrating AI data into clinical workflows can be time-consuming, especially if the tool doesn't align smoothly with existing electronic health records (EHR) or practice routines. Providers who work in clinical settings that prioritize efficient workflows may be more willing to learn about and utilize these data compared to providers in more resource-limited areas in which these tools may be viewed as disruptive or creating a burden to their workload.

Just as patient education will be important to promoting trust and acceptance, clinician education will be critical to promoting appropriate utilization of these tools [137]. Medical students have recognized the need for education about these tools in the medical curriculum [138–140]. Despite the perceived need for more education, no efforts have been reported to include AI education in medical curricula [137]. Somewhat ironically, the use of AI tools in education has been a major area of focus, including medical education. Many generalists do not have a high level of knowledge or experience with genetics in general, and PGx, specifically. Thus, their perceived confidence in their ability to evaluate the validity or relevance of data generated from an AI-PGx application may be further limited and contribute to reluctance to use these data for therapeutic decision-making. A lack of robust evidence for an AI-PGx tool's accuracy or outcomes may lead some providers to question its clinical utility. Roosan et al. developed an AI tool to educate providers about PGx called PGxKnow [141]. In addition, providers' comfort with digital tools and their level of technical proficiency can influence their adoption of AI-PGx in clinical settings. Provider acceptance may also be impacted by potential adverse impacts on the patient-provider discussed in the next section

Though different from the patient-provider relationship, similar impacts may be anticipated with the patient-pharmacist relationship. The role or placement of AI-assisted tools has also been explored in pharmacy settings. While the potential benefits of these tools have been recognized, pharmacists have expressed concern about patient privacy, impact on pharmacy practices, and legal issues [142]. And, similar to physicians, pharmacist knowledge and understanding of AI tools is limited [143].

### 3.4. Impact on patient-provider relationship

The introduction of AI raises questions about its potential impact on the fiduciary provider-patient relationship [119,144,145]. Providers and trainees have expressed a range of concerns about how the technology will impact their role as providers [138–140]. The inclusion of an AI 'third-party' can be either viewed as a welcome addition to the shared decision-making process or adversely impact the patient-provider relationship [146,147]. On the positive side, AI can enhance or validate a health provider's diagnosis or risk prediction, generating more patient confidence and potentially compliance with recommended interventions or medications. In addition, AI can potentially result in more patient-provider engagement since the provider will may have more time to focus on the patient that might otherwise be spent on reviewing a test report and determining the best treatment option [148]. AI tools can enhance provider empathy, relational behavior, and communication skills and a "reconceptualization of compassion as a *human-AI system* [149]."

On the negative side, however, AI could potentially diminish the relationship by reducing patient engagement and dehumanizing the patient-provider relationship [119,138,146]. Others have reported concerns of AI tools on patient autonomy, shared decision-making and trust in the patient-provider relationship [145]. AI may be perceived as infringing on providers' autonomy and/or patients' rights [119], with either the patient and provider deferring to the AI recommendation without critical evaluation of the personal circumstances or preferences of the patient. Patients may accord greater trust in a health provider's diagnosis and recommended interventions than a computer-based, detached analysis of their symptoms or test results. An over-reliance on AI tools may lead to feelings of patient disempowerment and reduced autonomy. The broader impact of replacing health providers' recommendations with AI-generated recommendations is potentially denial of specific interventions or medications. Furthermore, patients have expressed concerns regarding the replacement of provider empathy and compassion with an automated and digital-based delivery of healthcare and holistic care [112,150]. Transparency will be key to assigning (and/or accepting) the appropriate role of AI in clinical care and open engagement with patients.

The communication between patients and providers is critical to optimizing clinical decision-making and improved health outcomes. AI tools can fill gaps in provider access and enhance patient understanding. AI-generated patient materials have been reported equivalent or better than standard patient materials [151]. Patients have utilized AI tools like ChatGPT for health questions and diagnoses. In some cases, communication with AI tools can help patients gain better understanding of complex issues, enable repeated questioning of the AI tool and review of the responses, translation of responses, and sharing of responses with family members [152].

To date, there has not been as much written regarding the impact of AI on pharmacy practice and specifically the patient-pharmacist relationship despite the growing number of applications [153]. With pharmacists playing a major role in the delivery of PGx testing and test interpretation, some exploration of the impact of AI tools on the patient-pharmacist relationship will be needed.

### 3.5. Non-representativeness in AI-PGx datasets

AI models are trained on large datasets, but if the data are biased due to underrepresentation of certain patient populations, the AI system may yield biased or inaccurate results for some groups. Addressing biases in AI models is critical to avoid exacerbating health disparities [108,144]. This problem is not unique to PGx or AI research; several papers have demonstrated the extreme bias of study populations enrolled in genetic studies [154,155]. Effectively collecting and maintaining diverse datasets for AI-driven PGx research is crucial to ensure that AI models are representative and equitable. Using biased data sets in AI will continue to perpetuate the limited utility of these tools. Thus, datasets that represent the full spectrum of not only genetic diversity, but other patient variables (e.g., age, gender) are critical to ensure that AI models do not favor one group over others as has been demonstrated for some genetic tests [156].

One potential solution could ironically be linked to the advent of digital technologies and AI to reduce some of the barriers contributing to low participation rates of underrepresented populations. Specifically, remote data collection devices such as through mobile health apps and wearable devices could be used to enhance data collection from distant locations. In particular, wearables can reduce access barriers for geographically distant populations or older or disabled individuals who cannot easily travel and enable collection of a wider range of health data.

AI tools can also help facilitate open and ongoing engagement with study participants and promote informed decision-making to help address prospective participant concerns. Much like chatbots developed for clinical support, research-based chatbots can address prospective or enrolled participant concerns and questions. For genetic studies, in particular, concerns about data-sharing, access to samples, and other study risks and benefits can be answered through a chatbot assistant. Building trust through transparent and open communication about the goals, benefits, and privacy measures taken can help address these concerns and increase participation from under-represented groups. The novelty of AI will also warrant some background education for participants.

AI tools have also been explored to address clinical design issues, including recruitment [157,158]. For example, for multi-site trials, recruitment efficiency per site or attrition rates can be predicted and strategies adjusted to achieve recruitment goals. An AI-driven tool can facilitate eligibility screening [159,160]. A tool called Trial Pathfinder can enable adjustment of eligibility criteria for oncology trials [161]. Integrating AI-driven PGx tools with real-world evidence from EHRs and other clinical data sources will enable continuous learning and improvement of models.

### 3.6. Regulatory oversight of AI clinical applications

During this rapidly developing period of development and innovation, developing and implementing effective yet flexible oversight guidance and regulation is challenging. Establishing clear regulatory pathways and standards is necessary to ensure the safety and efficacy of AI-driven clinical tools [162]. AI-assisted tools are considered 'Software as a Medical Device (SaMD)' as outlined by the US Food and Drug Administration (FDA) [163]. The agency applies a 3-tier risk level framework (low, moderate, and high risk) in its review process.

Transparency and accountability are major challenges, not only for regulatory agencies [164], but for patient and providers as noted earlier. These challenges arise due to several factors, including the complexity of AI algorithms, the evolving nature of ML models, and creating a regulatory framework that balances innovation with patient safety. In some cases, it may not be completely clear how AI models make certain decisions. This lack of interpretability poses a major challenge for regulatory agencies when evaluating AI systems for clinical use as it is difficult to ensure the system's decisions are valid, fair, and safe for patient care. Requiring the use of explainable AI (XAI), where systems provide reasoning for their outputs is one possible approach. But many AI systems, particularly ML-based applications, are adaptive and therefore, present difficulties for continuous monitoring and post-market surveillance; an approved AI system could change after deployment, leading to new behaviors that were not part of the original evaluation. The FDA has introduced the concept of "total product lifecycle" regulation to address the circumstances presented by AI applications [165]. Navigating the regulatory oversight of this rapidly growing field is reminiscent of oversight efforts of early genomic profiles that generated risk predication scores based on an algorithm of gene expression values, referred to as *In Vitro* Diagnostic Multivariate Index Assays (IVDMIAs) [166]. Similar concerns were raised by the lack of transparency of how these scores were calculated for IVDMIAs. In summary. the "black-box" nature of many AI models limits their interpretability and presents challenges for patients and providers to understand, trust (and gain the trust of patients), and adopt these technologies. Enabling providers to understand how the tool works and specific qualities/metrics of the tool will be essential for clinical acceptance and routine utilization.

For AI tools, the variability in data quality and formats across different sources poses a significant challenge for oversight [74]. For example, variations in sequencing techniques, patient demographics, and use of unstructured text can lead to inconsistencies and affect model performance [167]. As of May 2024, the FDA has authorized 882 AI/ML-enabled medical devices [168]. Over the past decade, the FDA has released several discussion papers and guidance documents [169–171] as well as an agency inter-center report about oversight of AI tools. [172]. In a 2024 guidance, the FDA recognized the use of AI for this purpose a on the use of electronic health records (EHRs) or medical claims data in clinical studies as it pertains to regulatory decisions on drug safety or efficacy in clinical trials [173]. The European Medicines Agency has also been actively considering this field and issued a workplan in December 2023 to develop a collaborative and coordinated strategy [174].

Evaluating the success and reliability of AI-PGx models is critical to ensure their clinical application is both accurate and safe when determining how genetic variants influence drug responses. Specific metrics to assess the predictive power, robustness, and generalization ability of these models include accuracy, sensitivity and specificity, F-1 score (mean of precision and sensitivity; particularly useful for predicting rare outcomes), positive and negative values, and area under the curve-receiver operating characteristic (AUC-ROC). In particular, the AUC-ROC can help assess how well the model can differentiate between patients who will experience a beneficial or adverse reaction to a medication based on genetic markers. However, there remains a lack of standardization to apply and more effort is needed to inform research and development of AI tools for clinical use. Rethinking clinical trial designs for AI applications may be warranted to ascertain efficacy data.

## 4. Conclusion

In summary, addressing ELSI concerns in AI-driven PGx tools will be critical for the responsible development and use of these clinical applications. Ensuring that genetic data is anonymized or encrypted can help protect participant or patient privacy. Establishing strong data governance frameworks that outline the responsibilities and protocols for handling genetic data can help maintain data integrity and security.

Implementing clear and comprehensive informed consent processes can help ensure that participants/patients understand how their genetic data will be used, stored, and shared. Lastly, actively working to identify and mitigate biases in AI models is crucial to ensure equitable treatment and benefit across different patient populations. Therefore, it is critical to enroll diverse study populations or use diverse datasets for training AI models and continuously monitor and adjust models to address any detected biases.

More specifically related to the clinical setting, ensuring that AI models are transparent and their decision-making processes are explainable can build trust among both clinicians and patients. The novelty of these tools and shifting roles of provider and AI outputs may raise concerns among some patients and will warrant education and open communication to optimize utilization and patient confidence. Information about AI models should be provided to patients, including the benefits and risks of AI-driven PGx to enable them to make informed decisions about the use of AI for their healthcare. It will also be important to incorporate human oversight in the AI decision-making process. Clinicians and pharmacists can review AI-generated recommendations to ensure they are appropriate and add additional patient data to personalize the treatment guidance. Furthermore, creating mechanisms for continuous feedback from users, including clinicians and patients. Real-world feedback can help identify biases that may not have been evident during the initial development and testing. Training programs for health providers regarding the use of AI tools for therapeutic decision-making will help ensure appropriate integration into practice and can also raise awareness and provide tools and techniques for identifying and mitigating bias.

## 5. Future perspective

AI can significantly improve drug safety and health outcomes, leveraging its ability to analyze vast amounts of genomic and clinical data and detect patterns that are not feasible with traditional methods to inform therapeutic decision-making. The comprehensive analysis of PGx variant data with clinical measures can result in more precise drug selection and dosing to individual patients. The shifting roles in the provider-patient engagement will need to be addressed to minimize harms and maximize utilization of these tools. As PGx AI-tools are in the early phases of development and implementation, we should address potential ELSI in parallel to allow for the rapid transition to the clinical setting.

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